MA1522 Linear Algebra for Computing

Lesson 12: Row Space, Column Space, Nullspace and Rank

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson12` (e.g. `A1234567Z_lesson12.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 12**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Section 4.1 Column Space, Row Space, and Nullspace

Let A be an $m \times n$ matrix, written by columns and by rows as

$$A = \begin{pmatrix} | & | & & | \\ \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \\ | & | & & | \end{pmatrix} = \begin{pmatrix} — & \mathbf{r}_1 & — \\ & \vdots & \\ — & \mathbf{r}_m & — \end{pmatrix}.$$

Definition

1. **Row space**: $\text{Row}(A) = \text{span}\{\mathbf{r}_1, \ldots, \mathbf{r}_m\} \subseteq \mathbb{R}^n$.
2. **Column space**: $\text{Col}(A) = \text{span}\{\mathbf{c}_1, \ldots, \mathbf{c}_n\} \subseteq \mathbb{R}^m$.
3. **Nullspace**: $\text{Null}(A) = \{\, \mathbf{v} \in \mathbb{R}^n \mid A\mathbf{v} = \mathbf{0} \,\} \subseteq \mathbb{R}^n$, and $\text{nullity}(A) = \dim(\text{Null}(A))$.

Column space and consistency

$$\text{Col}(A) = \{\, A\mathbf{u} : \mathbf{u} \in \mathbb{R}^n \,\} = \{\, \mathbf{b} : A\mathbf{x} = \mathbf{b} \text{ is consistent} \,\}.$$

Section 4.1 Bases from the RREF

Suppose

$$A = \begin{pmatrix} | & & | \\ \mathbf{c}_1 & \cdots & \mathbf{c}_n \\ | & & | \end{pmatrix} = \begin{pmatrix} — & \mathbf{r}_1 & — \\ & \vdots & \\ — & \mathbf{r}_m & — \end{pmatrix} \xrightarrow{\text{rref}} R = \begin{pmatrix} | & & | \\ \mathbf{c}'_1 & \cdots & \mathbf{c}'_n \\ | & & | \end{pmatrix} = \begin{pmatrix} — & \mathbf{r}'_1 & — \\ & \vdots & \\ — & \mathbf{r}'_m & — \end{pmatrix}.$$

Theorem

1. Row operations preserve row space: $\text{Row}(A) = \text{Row}(R)$.
2. Row operations preserve **linear relations between columns**: $x_1\mathbf{c}_1 + \cdots + x_n\mathbf{c}_n = \mathbf{0}$ if and only if $x_1\mathbf{c}'_1 + \cdots + x_n\mathbf{c}'_n = \mathbf{0}$.

As a consequence, the subspaces $\text{Row}(A)$, $\text{Col}(A)$ and $\text{Null}(A)$ can be computed as follows.

(a) the nonzero $\mathbf{r}'_i$ form a basis for $\text{Row}(A)$;

(b) if $\mathbf{c}'_{i_1}, \ldots, \mathbf{c}'_{i_r}$ are the pivot columns of R, then $\mathbf{c}_{i_1}, \ldots, \mathbf{c}_{i_r}$ form a basis for $\text{Col}(A)$;

(c) $\text{Null}(A)$ is read off from the general solution of $R\mathbf{x} = \mathbf{0}$.

Question 1

Let $A = (\,\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3\ \mathbf{a}_4\ \mathbf{a}_5\,)$, where

$$A = \begin{pmatrix} 1 & -2 & 1 & 2 & 0 \\ 2 & -4 & 3 & 3 & 0 \\ 1 & -2 & 2 & 1 & 1 \\ -1 & 2 & 1 & -4 & -1 \end{pmatrix} \xrightarrow{\text{rref}} R = \begin{pmatrix} 1 & -2 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

(a) Find a basis for $\text{Row}(A)$.

(b) Find a basis for $\text{Col}(A)$, using columns of A.

(c) Find a basis for $\text{Null}(A)$.

Question 1 — solution

(a) The nonzero rows of R: $\{(1,-2,0,3,0),\ (0,0,1,-1,0),\ (0,0,0,0,1)\}$.

(b) The pivot columns of R are columns 1, 3, 5, so a basis for $\mathrm{Col}(A)$ is $\{\mathbf{a}_1, \mathbf{a}_3, \mathbf{a}_5\} = \{(1,2,1,-1)^T,\ (1,3,2,1)^T,\ (0,0,1,-1)^T\}$.

(c) Solve $R\mathbf{x} = \mathbf{0}$. The free variables are $x_2 = s$ and $x_4 = u$; then $x_1 = 2s - 3u$, $x_3 = u$, $x_5 = 0$:

$$\mathbf{x} = s \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + u \begin{pmatrix} -3 \\ 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \qquad \text{basis } \{(2,1,0,0,0)^T,\ (-3,0,1,1,0)^T\}.$$

□

Section 4.2 Rank

Let A be an $m \times n$ matrix.

Definition

The **rank** of A is $\text{rank}(A) = \dim(\text{Col}(A)) = \dim(\text{Row}(A))$, the number of pivot columns in the RREF of A. A is of **full rank** if $\text{rank}(A) = \min\{m, n\}$.

Rank-Nullity Theorem

$\text{rank}(A) + \text{nullity}(A) = n$, the number of columns of A.

Theorem (full rank equals number of rows)

The following are equivalent: (i) $\text{rank}(A) = m$; (ii) $\text{Col}(A) = \mathbb{R}^m$; (iii) the rows of A are linearly independent; (iv) $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^m$.

Question 2

Let A be an $m \times n$ matrix and B an $n \times p$ matrix. In this question we prove $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.

1. Show that
   (a) $\text{Col}(A) \supseteq \text{Col}(AB)$;
   (b) $\text{Null}(AB) \supseteq \text{Null}(B)$.

2. Show that $\text{rank}(AB) \leq \text{rank}(A)$ using (a), and $\text{rank}(AB) \leq \text{rank}(B)$ using (b).

Question 2 — solution

1(a) Let $\mathbf{y} \in \mathrm{Col}(AB)$. Then $\mathbf{y} = (AB)\mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^p$, so $\mathbf{y} = A(B\mathbf{x}) \in \{A\mathbf{u} : \mathbf{u} \in \mathbb{R}^n\} = \mathrm{Col}(A)$.

1(b) Let $\mathbf{v} \in \mathrm{Null}(B)$. Then $(AB)\mathbf{v} = A(B\mathbf{v}) = A\mathbf{0} = \mathbf{0}$, so $\mathbf{v} \in \mathrm{Null}(AB)$.

2. Take dimensions in (a):

$$\mathrm{rank}(AB) = \dim(\mathrm{Col}(AB)) \leq \dim(\mathrm{Col}(A)) = \mathrm{rank}(A).$$

Take dimensions in (b): $\mathrm{nullity}(B) \leq \mathrm{nullity}(AB)$. Both B and AB have p columns, so by Rank-Nullity

$$p - \mathrm{rank}(B) \leq p - \mathrm{rank}(AB), \qquad \text{that is,} \qquad \mathrm{rank}(AB) \leq \mathrm{rank}(B).$$

□

Question 3

Let A be an $m \times n$ matrix.

(a) Show that $\text{Null}(A^T A) = \text{Null}(A)$.

(b) Suppose that A is of full rank and $m \geq n$. Show that $A^T A$ is invertible, and that $(A^T A)^{-1} A^T$ is a left inverse of A.

Question 3 — solution: (a)

($\supseteq$) If $A\mathbf{v} = \mathbf{0}$, then $(A^TA)\mathbf{v} = A^T(A\mathbf{v}) = A^T\mathbf{0} = \mathbf{0}$. (This is Question 2, part 1(b), for the product A^TA.)

($\subseteq$) Let $A^TA\mathbf{v} = \mathbf{0}$ and write $A\mathbf{v} = \mathbf{w} = (w_1, \ldots, w_m)^T$. Multiply on the left by $\mathbf{v}^T$:

$$0 = \mathbf{v}^TA^TA\mathbf{v} = (A\mathbf{v})^T(A\mathbf{v}) = \mathbf{w}^T\mathbf{w} = w_1^2 + w_2^2 + \cdots + w_m^2.$$

Each w_i is real, so $w_1 = \cdots = w_m = 0$, that is, $A\mathbf{v} = \mathbf{0}$ and $\mathbf{v} \in \text{Null}(A)$.

Hence $\text{Null}(A^TA) = \text{Null}(A)$.

Question 3 — solution: (b)

Since $m \geq n$, full rank means $\text{rank}(A) = \min\{m, n\} = n$. By Rank-Nullity, $\text{nullity}(A) = n - n = 0$, so $\text{Null}(A) = \{\mathbf{0}\}$.

By (a), $\text{Null}(A^T A) = \text{Null}(A) = \{\mathbf{0}\}$: the system $(A^T A)\mathbf{x} = \mathbf{0}$ has only the trivial solution. Since $A^T A$ is a square matrix of order n, it is invertible.

Finally,

$$\big((A^T A)^{-1} A^T\big) A = (A^T A)^{-1}(A^T A) = I_n,$$

so $(A^T A)^{-1} A^T$ is a left inverse of A. □

Next lesson

- **Recess Week**: Sat 19 – Sun 27 September. No class.
- Next lesson: **Tue 29 Sep**, **Sections 5.1 & 5.2 Orthogonality, and Orthogonal and Orthonormal Bases**. Please watch the video clips beforehand.